

Steven M. Goodman, Ph.D.

smgoodmn [at] gmail [dot] com · stevenmgoodman.com

Curriculum Vitae (August 2025)

About Me

I am a User Researcher, Designer, and Technologist with seven years of experience leading mixed-methods initiatives to understand user needs, surface actionable product recommendations, and build systems that work for end users. I have a Ph.D. in Human Centered Design & Engineering from the University of Washington, and my publications have been awarded recognition at top human-computer interaction (HCI) venues like CHI and ASSETS. My work in the University of Washington's Inclusive Design Lab and Google Research examined the intersection of human-centered AI and accessibility, exploring topics like personalizable sound recognition and AI-enhanced writing support tools. Previously, I supported the development of wearable systems at NASA and the University of Minnesota.

I specialize in designing and evaluating AI/ML systems (LLMs, generative AI, predictive models) to ensure they are understandable, intuitive, and useful. I also have deep expertise conducting research with and designing for specialized user groups, including people with disabilities. My experience spans the entire project lifecycle, from generative research and journey mapping to high-fidelity prototyping and usability testing.

Education

2018 - 2024	DOCTOR OF PHILOSOPHY in Human Centered Design & Engineering University of Washington, Seattle, WA Dissertation: <i>"Human-Centered Sound Recognition Tools for Deaf and Hard of Hearing Users"</i> Advisor: Prof. Leah Findlater
2018 - 2024	MASTER OF SCIENCE in Human Centered Design & Engineering University of Washington, Seattle, WA
2014 - 2018	BACHELOR OF SCIENCE in Mathematics University of Minnesota, Minneapolis, MN

Professional Experience

Dec. 2024 - Present	SENIOR AI STRATEGIST Range Digital Minneapolis, MN (Remote) – Advise digital marketing agency on opportunities to integrate AI/ML within existing business workflows, leveraging expertise in user research and predictive and generative models. – Includes prototyping generative AI content tools and testing customer lifetime value (LTV) models.
---------------------	--

Sept. 2018 - Dec. 2024	GRADUATE USER RESEARCHER, Inclusive Design Lab University of Washington Seattle, WA – Led multi-year research initiative to understand the needs of specialized user groups when engaging with data-driven technology, yielding design frameworks that guided the development of 4 AI-driven assistive technology prototypes. – Conducted end-to-end research cycles using mixed methods incl. participatory design, contextual inquiry, diary studies, interviews, and usability studies to understand user behaviors, motivations, and pain points. Interviewed & tested w/ 50+ participants. – Built and evaluated functional web and mobile prototypes for user testing, leading to 12 peer-reviewed publications (3 as lead author) at top HCI venues, including CHI, IMWUT, and ASSETS. – Managed and collaborated with a multidisciplinary research team of students and faculty, mentoring 5 junior team members and delegating responsibilities to achieve research objectives.
Sept. 2021 - April 2022	RESEARCH INTERN, People + AI Research (PAIR) Google Seattle, WA – Led effort exploring early (pre-ChatGPT) large language models (LLMs) as productivity and accessibility tools, owning the research plan, prototype development, execution of the user study, analysis, and reporting, working in collaboration with a large, cross-functional team. – Designed and built “LaMPost” (TypeScript, Node.js), a high-fidelity prototype application, translating user needs from formative work to the design of AI support features (summarize, rewrite, suggest). – Conducted mixed-methods study with 19 participants to assess the prototype’s value and usability, employing a think-aloud protocol, semi-structured interviews, and rating scales. – Synthesized user data into a final report identifying high-value features (rewriting, subject lines) and critical issues (model accuracy, choice overload), influencing product roadmap for LLM applications and leading to two peer-reviewed publications (ASSETS, CACM).
June 2015 - May 2018	RESEARCH ASSISTANT, Wearable Technology Lab University of Minnesota Minneapolis, MN – Co-developed novel design and manufacture process for electronic textiles based on PCB designs, leading to Honorable Mention at ISWC 2017.
Summer 2017	HUMAN FACTORS INTERN, Space Suit Assembly Team NASA Johnson Space Center Houston, TX – Conducted human factors evaluation to assess operational effort of a next-generation spacesuit, collecting and analyzing quantitative performance data.
Summer 2016	INTERN, Wearable Electronics Application & Research Lab NASA Johnson Space Center Houston, TX – Redesigned housing for wearable CO2 monitor to improve usability, integrate new hardware, and streamline assembly for use in microgravity

Publications

- | | |
|------|---|
| 2025 | ¹⁷ SPECTRA: PERSONALIZABLE SOUND RECOGNITION FOR DEAF AND HARD OF HEARING USERS THROUGH INTERACTIVE MACHINE LEARNING
<i>Steven Goodman</i> , Emma McDonnell, Jon E. Froehlich, Leah Findlater
ACM CHI 2025 (PDF Talk Demo Github doi) |
| 2024 | ¹⁶ HUMAN-CENTERED SOUND RECOGNITION TOOLS FOR DEAF AND HARD OF HEARING USERS
<i>Steven Goodman</i>
Doctoral Dissertation, University of Washington (PDF ProQuest) |

- 2024 15 **LAMPOST: AI WRITING ASSISTANCE FOR ADULTS WITH DYSLEXIA USING LARGE LANGUAGE MODELS**
Steven Goodman, Erin Buehler, Patrick Clary, Andy Coenen, Aaron Donsbach, Tiffanie N. Horne, Michal Lahav, Robert MacDonald, Rain Breaw Michaels, Ajit Narayanan, Mahima Pushkarna, Joel Riley, Alex Santana, Lei Shi, Rachel Sweeney, Phil Weaver, Ann Yuan, Meredith Ringel Morris
Communications of the ACM Vol. 67, No. 9, Invited Article ([PDF](#) | [CACM](#) | [doi](#))
- 2023 14 **“EASIER OR HARDER, DEPENDING ON WHO THE HEARING PERSON IS”: CODESIGNING VIDEOCONFERENCING TOOLS FOR SMALL GROUPS WITH MIXED HEARING STATUS**
 Emma McDonnell, Soo Hyun Moon, Lucy Jiang, *Steven Goodman*, Raja Kushalnagar, Jon E. Froehlich, Leah Findlater
ACM CHI 2023 ([PDF](#) | [doi](#))
- 2022 13 **LAMPOST: DESIGN AND EVALUATION OF AN AI-ASSISTED EMAIL WRITING PROTOTYPE FOR ADULTS WITH DYSLEXIA**
Steven Goodman, Erin Buehler, Patrick Clary, Andy Coenen, Aaron Donsbach, Tiffanie N. Horne, Michal Lahav, Robert MacDonald, Rain Breaw Michaels, Ajit Narayanan, Mahima Pushkarna, Joel Riley, Alex Santana, Lei Shi, Rachel Sweeney, Phil Weaver, Ann Yuan, Meredith Ringel Morris
ACM ASSETS 2022, Best Paper Honorable Mention ([PDF](#) | [Talk](#) | [doi](#))
- 12 **SOUNDWATCH: DEEP LEARNING FOR SOUND ACCESSIBILITY ON SMARTWATCHES**
 Dhruv Jain, Hung Ngo, Pratyush Patel, *Steven Goodman*, Khoa Nguyen, Rachel Grossman-Kahn, Leah Findlater, Jon Froehlich
Communications of the ACM Vol. 65, No. 6, Invited Article ([PDF](#) | [CACM](#) | [doi](#))
- 11 **PROTOSOUND: A PERSONALIZED, SCALABLE SOUND RECOGNITION SYSTEM FOR D/DEAF AND HARD-OF-HEARING USERS**
 Dhruv Jain, Khoa Nguyen, *Steven Goodman*, Rachel Grossman-Kahn, Hung Ngo, Aditya Kusupati, Ruofei Du, Alex Olwal, Leah Findlater, Jon Froehlich
ACM CHI 2022 ([PDF](#) | [Demo](#) | [doi](#))
- 2021 10 **TOWARD USER-DRIVEN SOUND RECOGNIZER PERSONALIZATION WITH PEOPLE WHO ARE DEAF OR HARD OF HEARING**
Steven Goodman, Ping Liu, Dhruv Jain, Emma J. McDonnell, Jon Froehlich, Leah Findlater
ACM IMWUT 2021 ([PDF](#) | [Talk](#) | [doi](#))
- 9 **SOCIAL, ENVIRONMENTAL, AND TECHNICAL: FACTORS AT PLAY IN THE CURRENT USE AND FUTURE DESIGN OF SMALL-GROUP CAPTIONING**
 Emma McDonnell, Ping Liu, *Steven Goodman*, Raja Kushalnagar, Jon Froehlich, Leah Findlater
PACMHCI CSCW 2021, Best Paper Honorable Mention ([PDF](#) | [doi](#))
- 2020 8 **EVALUATING SMARTWATCH-BASED SOUND FEEDBACK FOR DEAF AND HARD-OF-HEARING USERS ACROSS CONTEXTS**
Steven Goodman, Susanne Kirchner, Rose Guttman, Dhruv Jain, Jon Froehlich, Leah Findlater
ACM CHI 2020 ([PDF](#) | [doi](#))

- | | |
|------|--|
| | <p>7 SOUNDWATCH: EXPLORING SMARTWATCH-BASED DEEP LEARNING APPROACHES TO SUPPORT SOUND AWARENESS FOR DEAF AND HARD OF HEARING USERS</p> <p>Dhruv Jain, Hung Ngo, Pratyush Patel, <i>Steven Goodman</i>, Leah Findlater, Jon Froehlich</p> <p>ACM ASSETS 2020, <i>Best Artifact Award</i> (PDF doi)</p> |
| | <p>6 HOLOSOUND: COMBINING SPEECH AND SOUND IDENTIFICATION FOR DEAF OR HARD OF HEARING USERS ON A HEAD-MOUNTED DISPLAY</p> <p>Ru Guo, Robin Yiru Yang, Johnson Kuang, Xue Bin, Dhruv Jain, <i>Steven Goodman</i>, Leah Findlater, Jon Froehlich</p> <p>ACM ASSETS 2020 (PDF Demo doi)</p> |
| | <p>5 FIELD STUDY OF A TACTILE SOUND AWARENESS DEVICE FOR DEAF AND HARD OF HEARING USERS</p> <p>Dhruv Jain, Brendon Chiu, <i>Steven Goodman</i>, Chris Schmandt, Leah Findlater, Jon Froehlich</p> <p>ACM ISWC 2020 (PDF doi)</p> |
| | <p>4 HOMESOUND: AN ITERATIVE FIELD DEPLOYMENT OF AN IN-HOME SOUND AWARENESS SYSTEM FOR DEAF OR HARD OF HEARING USERS</p> <p>Dhruv Jain, Kelly Mack, Akli Amrous, <i>Steven Goodman</i>, Matt Wright, Leah Findlater, Jon Froehlich</p> <p>ACM CHI 2020 (PDF doi)</p> |
| 2019 | <p>3 SOCIAL TENSIONS WITH HEAD-MOUNTED DISPLAYS FOR ACCESSIBILITY</p> <p><i>Steven Goodman</i>, Dhruv Jain, Jon Froehlich, Brock Craft, Leah Findlater</p> <p>ACM CHI 2019, Social HMD Workshop (PDF)</p> |
| | <p>2 FAIRNESS ISSUES IN AI SYSTEMS THAT AUGMENT SENSORY ABILITIES</p> <p>Leah Findlater, <i>Steven Goodman</i>, Yuhang Zhao, Shiri Azenkot, Margot Hanley</p> <p>ACM SIGACCESS Accessibility and Computing, Oct 2019, Issue 125 (PDF doi)</p> |
| 2017 | <p>1 SURFACE-MOUNT MANUFACTURING FOR E-TEXTILE CIRCUITS</p> <p>Md. Tahmidul Islam Molla, <i>Steven Goodman</i>, Nicholas Schleif, Mary Ellen Berglund, Cade Zacharias, Crystal Compton, Lucy E. Dunne</p> <p>ACM ISWC 2017, <i>Best Paper Honorable Mention (top 3% of submissions)</i> (doi)</p> |

Teaching Experience

- | | |
|-------------|---|
| Spring 2020 | <p>TEACHING ASSISTANT, Accessibility and Inclusive Design (HCDE 598A)</p> <p>Dept. of Human Centered Design and Engineering, University of Washington, Seattle, WA</p> <p>Instructor: Dr. Leah Findlater</p> |
| Fall 2019 | <p>TEACHING ASSISTANT, Interactive Systems Design and Technology (HCDE 310A)</p> <p>Dept. of Human Centered Design and Engineering, University of Washington, Seattle, WA</p> <p>Instructor: Dr. Sean Munson</p> |

Selected Awards and Honors

2024	INVITED RESEARCH HIGHLIGHT, Communications of the Association for Computing Machinery Goodman, Buehler, Clary, Coenen, Donsbach, Horne, Lahav, MacDonald, Michaels, Narayanan, Pushkarna, Riley, Santana, Shi, Sweeney, Weaver, Yuan, Morris. “LaMPPost: AI Writing Assistance for Adults with Dyslexia Using Large Language Models” (CACM Technical Perspective) Work recognized for its broad impact and significance to the computing field by the flagship publication of the ACM.
2022	HONORABLE MENTION, 2022 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) Goodman, Buehler, Clary, Coenen, Donsbach, Horne, Lahav, MacDonald, Michaels, Narayanan, Pushkarna, Riley, Santana, Shi, Sweeney, Weaver, Yuan, Morris. “LaMPPost: Design and Evaluation of an AI-assisted Email Writing Prototype for Adults with Dyslexia”
2020	GRADUATE RESEARCH FELLOWSHIP, National Science Foundation (est. \$138,000) <i>NSF GRFP</i>. Awarded to top graduate student applicants in NSF-supported STEM fields. Provides financial support in the form of a stipend and tuition waiver. BEST ARTIFACT AWARD, 2020 ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) Jain, Ngo, Patel, Goodman, Findlater, Froehlich. “SoundWatch: Exploring Smartwatch-based Deep Learning Approaches to Support Sound Awareness for Deaf and Hard of Hearing Users.” (Forbes Yahoo News UW News)
2019	RUNNER UP, Madrona Prize Jain, Mack, Goodman, Findlater, Froehlich. “HomeSound: Exploring Sound Awareness in the Home for People Who Are Deaf and Hard of Hearing.” University of Washington. (Bloomberg GeekWire)
2018	UNDERGRADUATE RESEARCH OPPORTUNITIES GRANT, University of Minnesota (\$1,800) Goodman, Dunne. “Haptic Feedback Garments for Visual Accessibility.”
2017	HONORABLE MENTION, 2017 ACM International Symposium on Wearable Computers (ISWC) Molla, Goodman, Schleif, Berglund, Zacharias, Compton, Dunne. “Surface-Mount Manufacturing for E-Textile Circuits.” Top 3% of submissions.
2014 - 2018	SCHOLARSHIPS, University of Minnesota Annual merit-based awards from the Tozer Foundation (\$10,000), A. & A. Berggren (\$8,000), Lemberg Engineering (\$4,000).